		$-V_i$
2	**A**	For the upward motion, air resistance and gravitational force are both acting downwards, acceleration is greater than the acceleration of free fall and hence it decelerates faster to reach the maximum height. As for the downward motion, air resistance and gravitational force are in opposite directions and the resultant force equals to gravitational force minus the air resistance. Acceleration is smaller than the acceleration of free fall and hence it accelerates slower downwards than it decelerates upwards.
3	**D**	Consider cart 3 and cart 4 as single object of mass 800 kg